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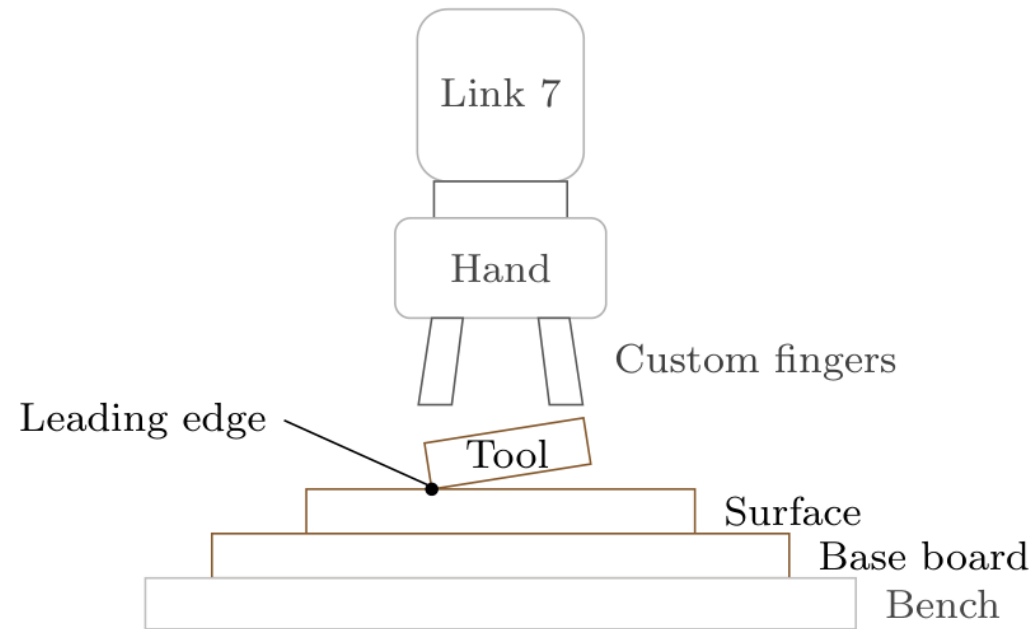
Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

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- Problem

- Grinding requires an aligned tool face.
- Unknown orientation mismatch between configured and physical surfaces.

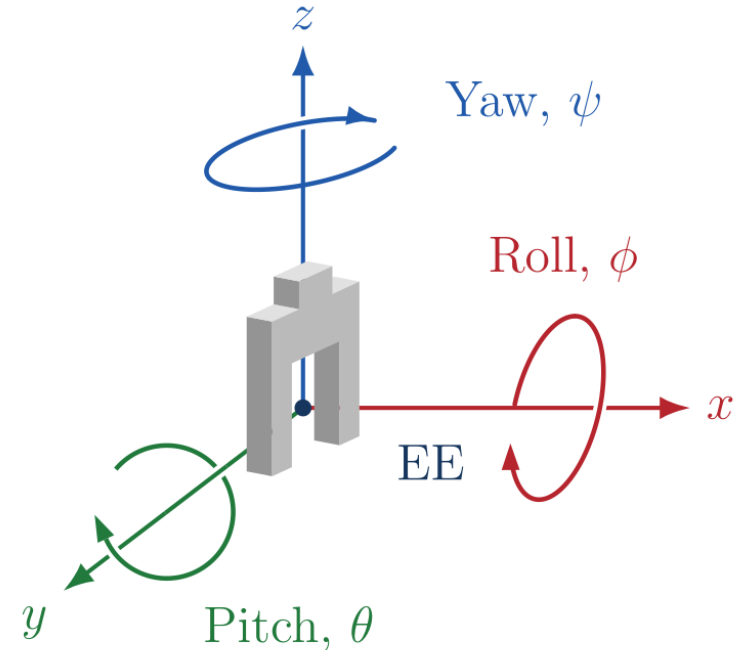
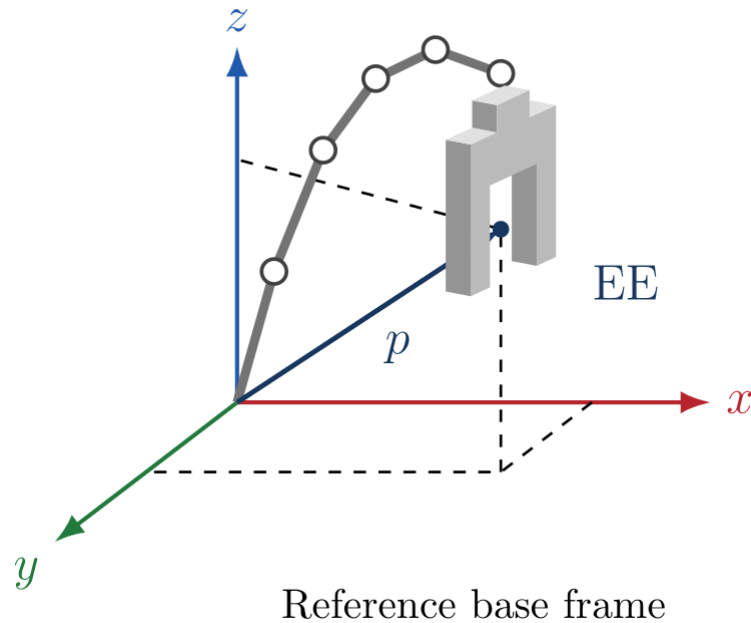
- Idea

- Cartesian impedance makes position and rotation compliant.
- Pressing a tilted tool creates a contact moment.
- Rotational compliance allows the tool to turn towards alignment.

1. Introduction
2. Controller design
3. Experimental methodology
4. Validation
5. Experimental results
6. Conclusion and future work

Cartesian pose

- The end effector (EE) is the tool or gripper at the end of the robot arm.
- Cartesian pose describes its position and rotation in 3D (6 degrees of freedom).
 - Position: 3 translations
 - Orientation: 3 rotations



- The EE position depends on the configuration of all 7 joint angles: $p = p(q)$

- Position correction by force

$$e_p = p_d - p_{EE}$$

$$f = K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE})$$

- Orientation correction by moment

$$e_R = \phi u, \quad R_d R_{EE}^\top = R(u, \phi)$$

$$m = K_R e_R + D_R (\omega_d - \omega_{EE})$$

- Cartesian wrench

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{\begin{bmatrix} K_p & \mathbf{0} \\ \mathbf{0} & K_R \end{bmatrix}}_{K_c} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{\begin{bmatrix} D_p & \mathbf{0} \\ \mathbf{0} & D_R \end{bmatrix}}_{D_c} \begin{bmatrix} \dot{p}_d - \dot{p}_{EE} \\ \omega_d - \omega_{EE} \end{bmatrix}$$

- Zero off-diagonal blocks: translation and rotation are decoupled.

K_p, K_R translational / rotational stiffness

D_p, D_R translational / rotational damping

$\dot{p}, \omega$ linear / angular velocity

d, EE desired / measured end effector

- Joint-torque commands from Cartesian wrench

$$F = \begin{bmatrix} f \\ m \end{bmatrix}, \quad \tau_{\text{cart}} = J^{\top}(q)F$$

- Jacobian $J(q)$: joint velocities $\rightarrow$ EE linear and angular velocities



1 kHz control loop · C++ / libfranka

- Impedance reference point

Tool centre point (TCP): controlled reference point on the EE

p_c Virtual centre of compliance (CoC) position

$r_c = p_c - p_{\text{TCP}}$ Displacement from TCP to CoC

$A = \text{Ad}(r_c)$ Impedance reference: TCP $\rightarrow$ CoC

- Stiffness

$$K_{\text{TCP}} = A^\top K_c A$$

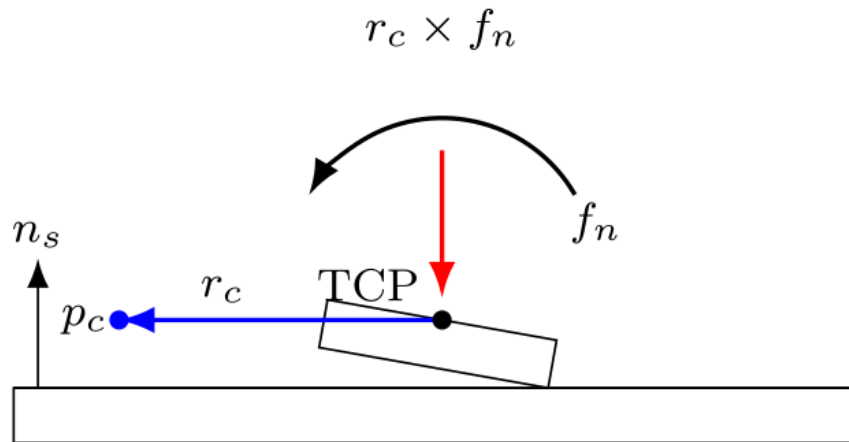
- Damping

$$D_{\text{TCP}} = A^\top D_c A$$

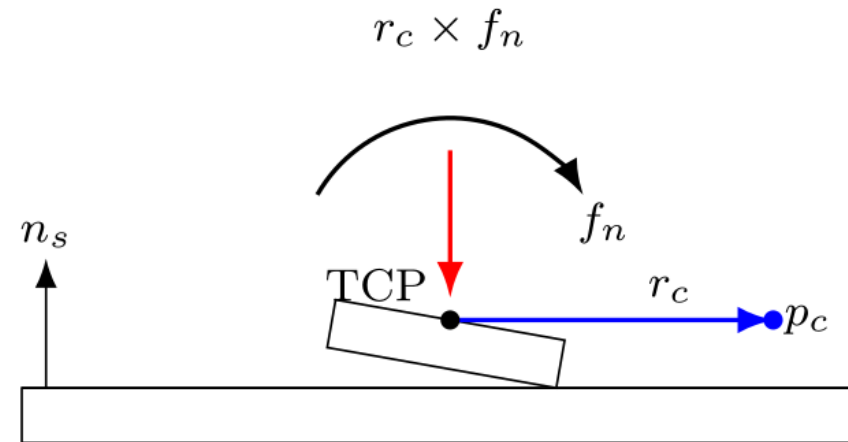
- Non-zero off-diagonal blocks in K and D couple translation and rotation.
- CoC displacement links the commanded force f and moment m .

Centre of compliance (CoC)

- Effect of CoC displacement
- Added moment can support or oppose rotation.
- Commanded force and moment can change.



(a) Supporting displacement



(b) Reversed displacement

- Commanded TCP moment

$$m = m_R + r_c \times f$$

m_R rotational spring-damper moment

- The commanded force acts as if applied at the virtual point p_c .
- Opposite displacement reverses the added moment.

Null-space controller

- Use extra joint freedom to manage the arm's disturbance response.

- **Redundancy**

7 joints for a 6-DOF Cartesian pose.
One redundant direction at full rank.

$$\dim(\ker J) = 7 - 6 = 1$$

- **Cartesian task preservation**

Zero instantaneous TCP translation and rotation

$$J(q) \dot{q}_{\text{null}} = 0$$

- **Damping** τ_d

Opposes redundant joint velocity.
Suppresses unwanted joint motion.

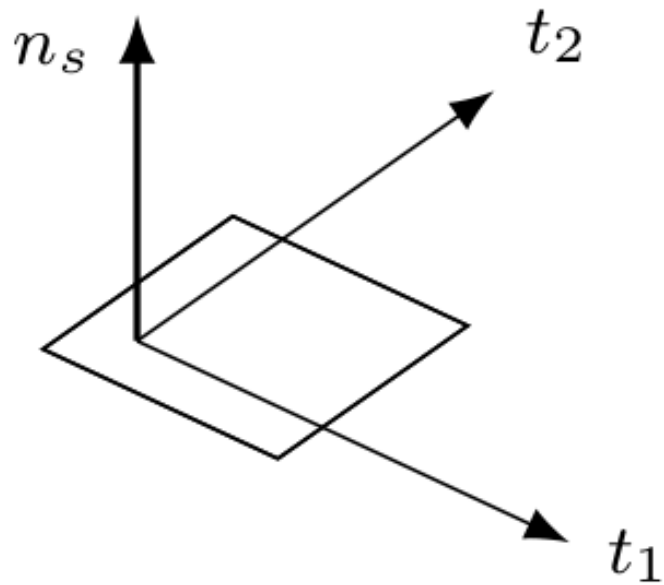
- **Conditioning** τ_σ

Seeks easier motion in the weakest tool direction.
Can initiate joint motion.

- Singularity: the EE loses at least one instantaneous motion direction.
- Near a singularity, some EE motions need very large joint velocities.

Tool orientation → Surface approach → Contact establishment → Surface grinding

- Surface-relative compliance



n_s outward surface normal

t_1, t_2 perpendicular surface tangents

- Compliant contact settings

- Normal translation

$$K_{p,n} = 350 \text{ N/m}$$

- Tangential translation

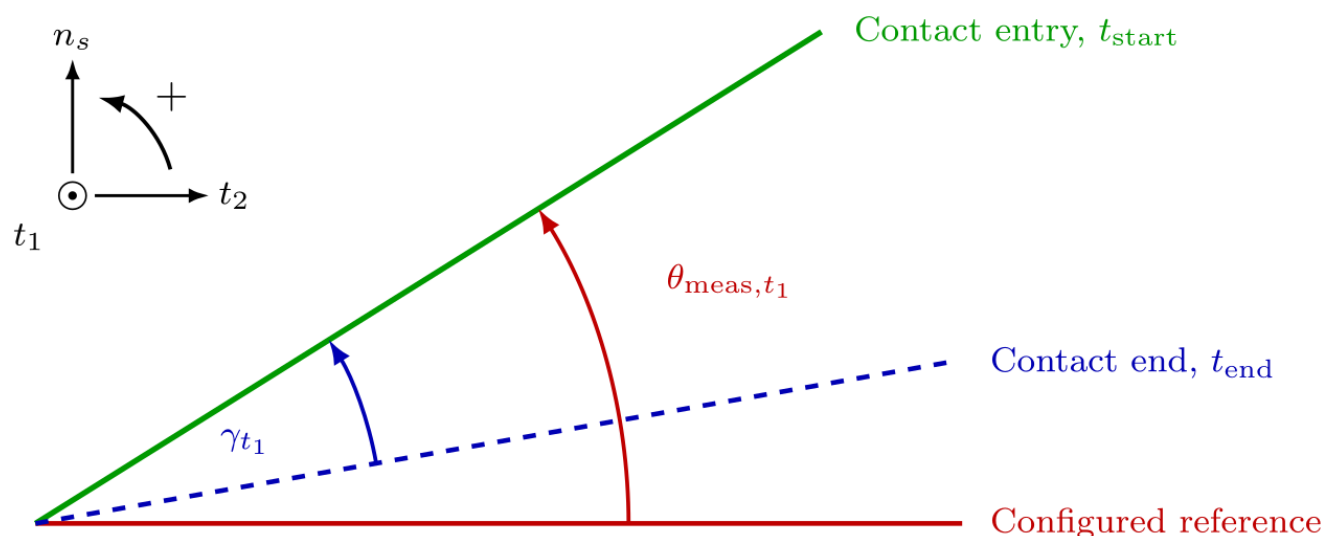
$$K_{p,t_1} = K_{p,t_2} = 2000 \text{ N/m}$$

- Rotation about the surface normal

$$K_{R,n} = 50 \text{ N m/rad}$$

- Rotations about the surface tangents

$$K_{R,t_1} = K_{R,t_2} = 5 \text{ N m/rad}$$



θ_{meas, t_1} Measured angular offset
Entry tilt relative to the
configured surface

γ_{t_1} Contact response
End-to-entry EE rotation
about the first tangent

- Aligning response: same sign as the measured angular offset
- Response sign is opposite to the EE's start-to-end rotation
- EE rotation is measured. Physical alignment is assessed indirectly.

- Commanded joint disturbance

$$\tau_{\text{dist}} = J_p(q)^\top f_{\text{dist}}$$

Joint-torque equivalent of a virtual 20 N point force on link 3.

- Four settings with three trials each
 - No null-space torque
 - Damping: 2 N m s/rad
 - Disturbance interval: 5–9 s, displayed as 0–4 s.

- Cartesian pose hold

Hold the captured position and orientation.

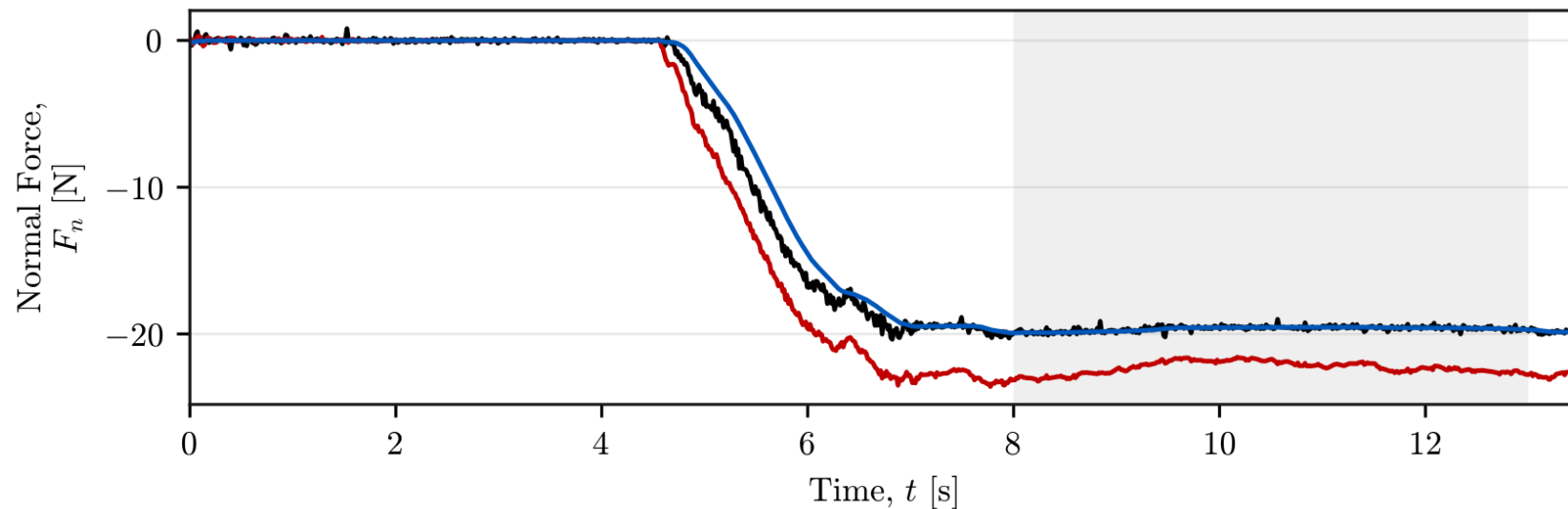
- Conditioning active for 5 s before disturbance onset
- Conditioning: 1.5 and 2.0 N m
- Total motion, conditioning and net displacement

Normal-force plausibility assessment

- Manual push: about 20 mm

Measured error change: -19.650 mm

$$\begin{aligned} F_{n,qs} &= K_{p,n} n_s^\top (e_p - e_{p,0}) \\ &= (1000 \text{ N/m}) \cdot (-0.019650 \text{ m}) = -19.650 \text{ N} \end{aligned}$$



— Commanded

— Model-estimated

— Quasi-static prediction

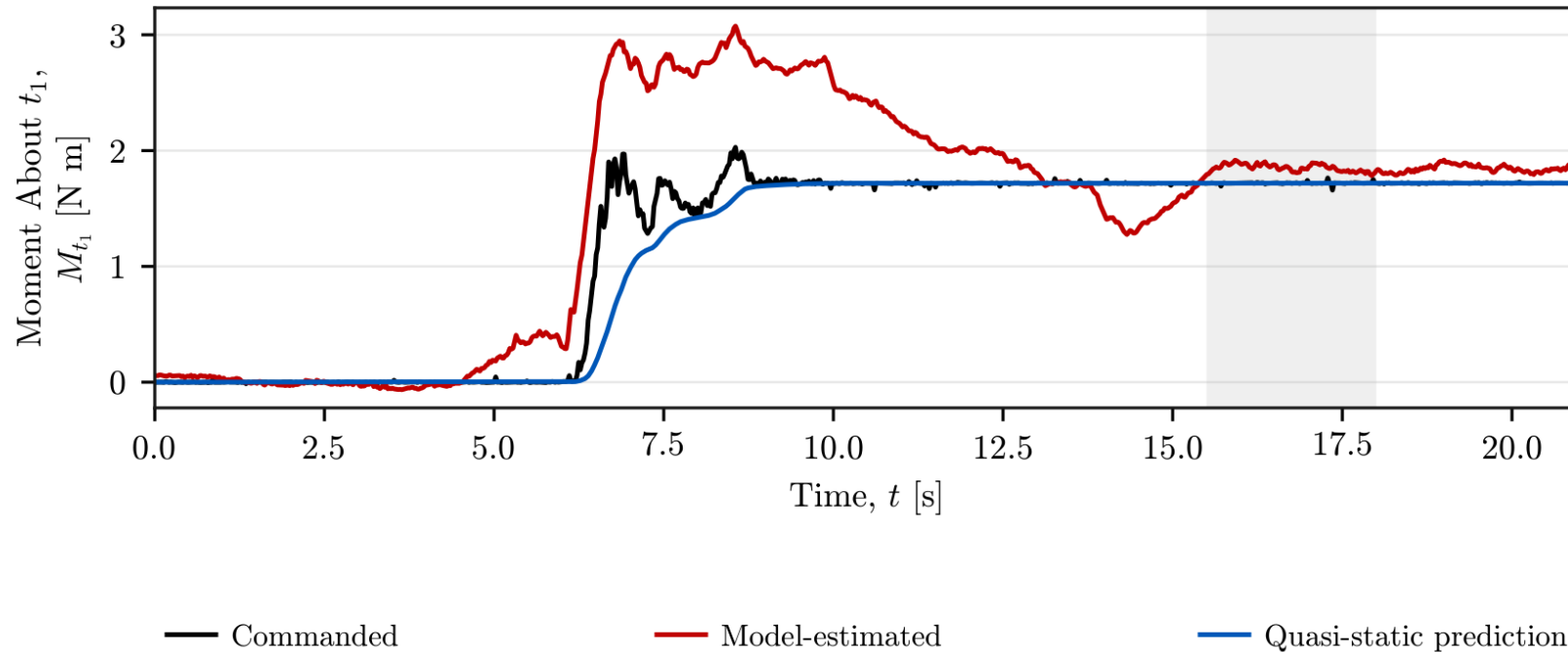
- Commanded mean: -19.640 N (0.05% from calculation)
- Model-estimated mean: -22.292 N (13.50% from command)

Moment plausibility assessment

- Manual rotation: about 6.5°

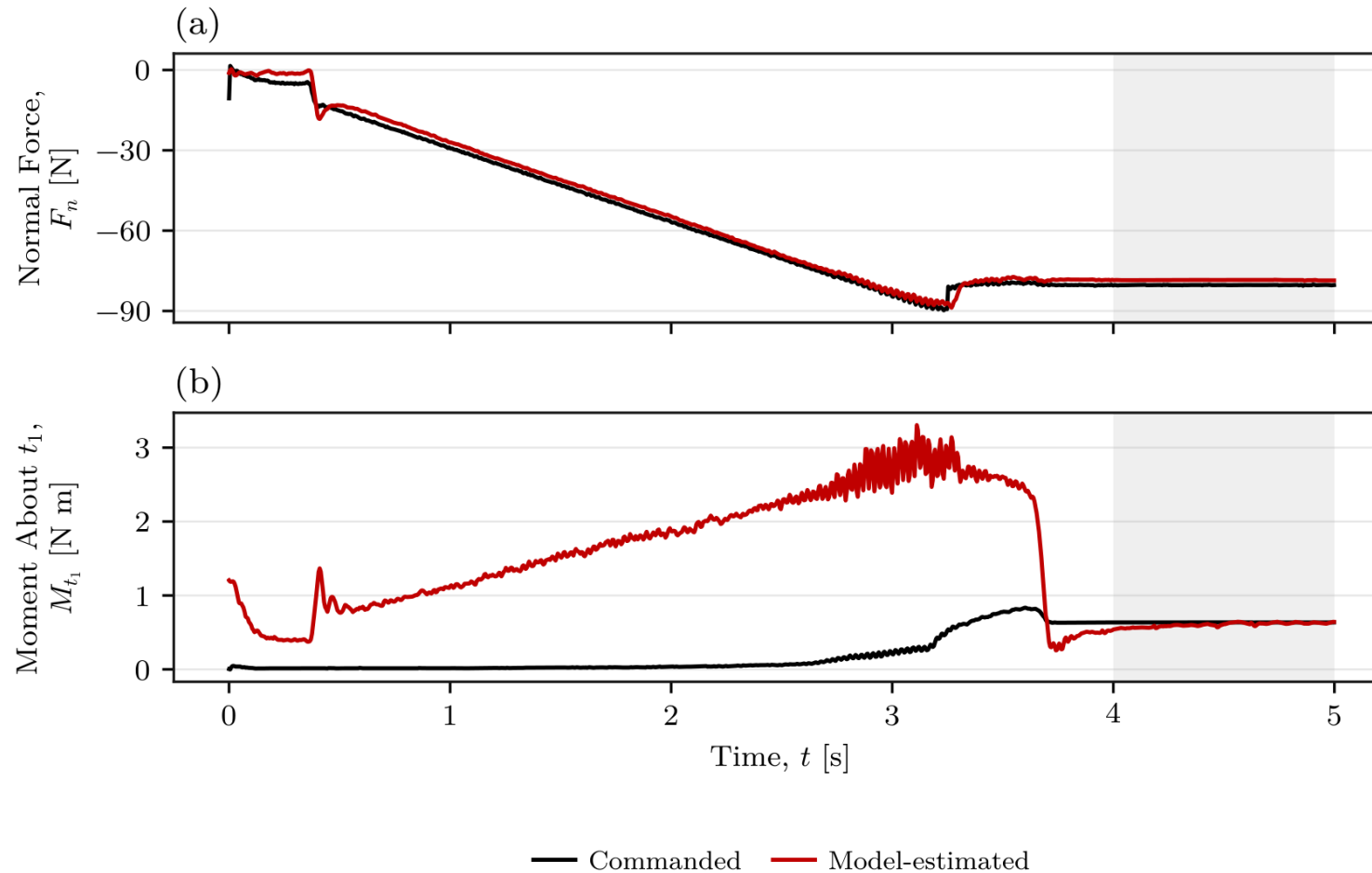
Measured error change: 6.565°

$$\begin{aligned} M_{t_1, \text{qs}} &= K_{R, t_1} t_1^\top (e_R - e_{R, 0}) \\ &= (15 \text{ N m/rad}) \cdot \left(6.565 \cdot \frac{\pi}{180} \text{ rad} \right) \approx 1.719 \text{ N m} \end{aligned}$$



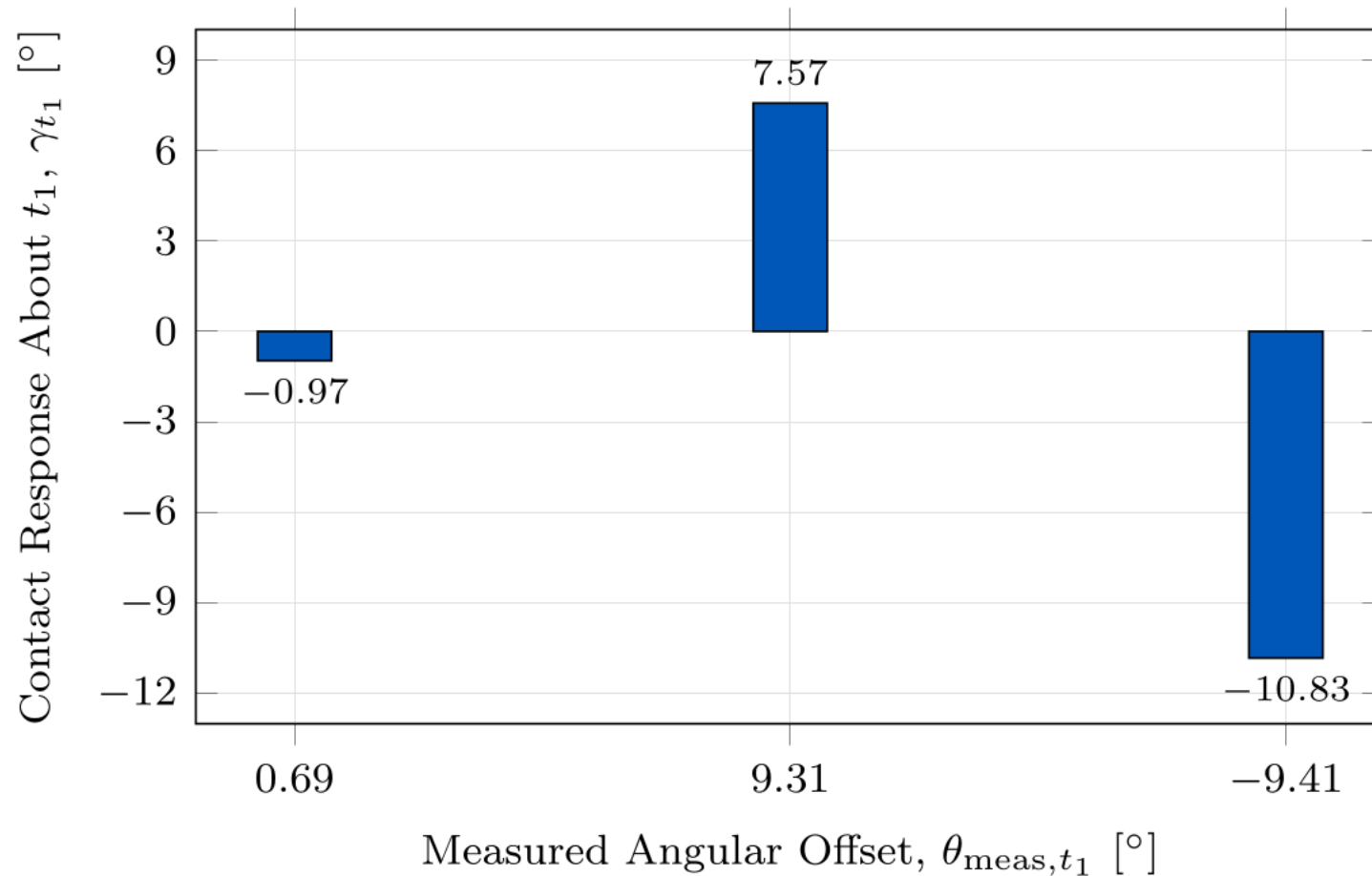
- Commanded mean: 1.719 N m (matches calculation)
- Model-estimated mean: 1.852 N m (7.75% from command)

Commanded and estimated wrench



- Stationary force difference: 2.32%
- Stationary moment difference: 4.87%
- Separate trial during contact establishment

Baseline contact response

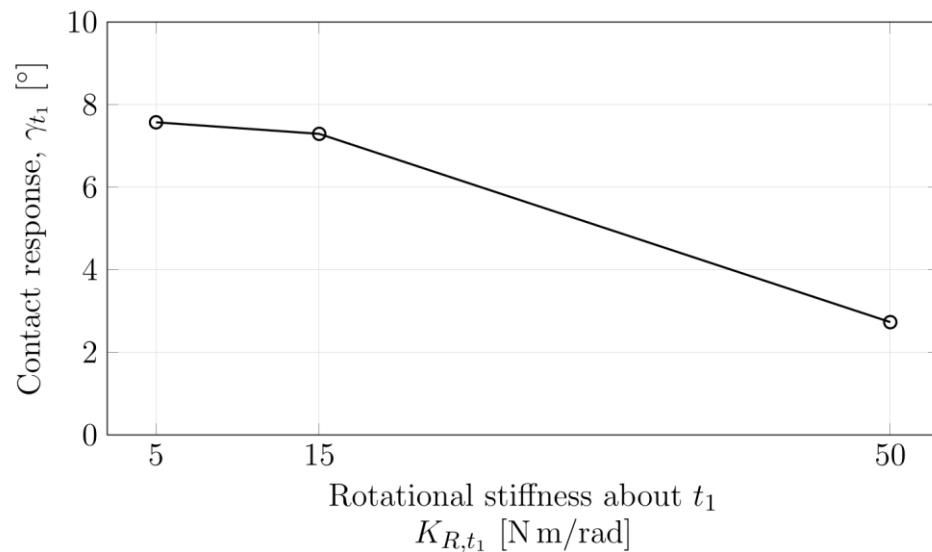


- Centre of compliance at the TCP
- EE rotation towards the configured surface
- Physical alignment assessed indirectly

Effect of stiffness

- Rotational stiffness

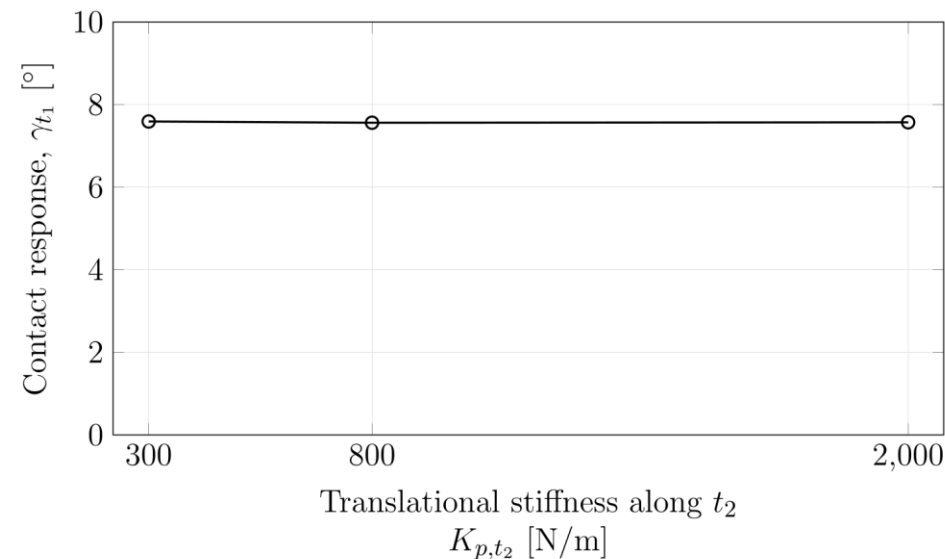
Resistance to rotation about t_1



Measured angular offset
 $\theta_{\text{meas},t_1} = (9.31\text{--}9.32)^\circ$

- Tangential stiffness

Resistance to sliding along t_2

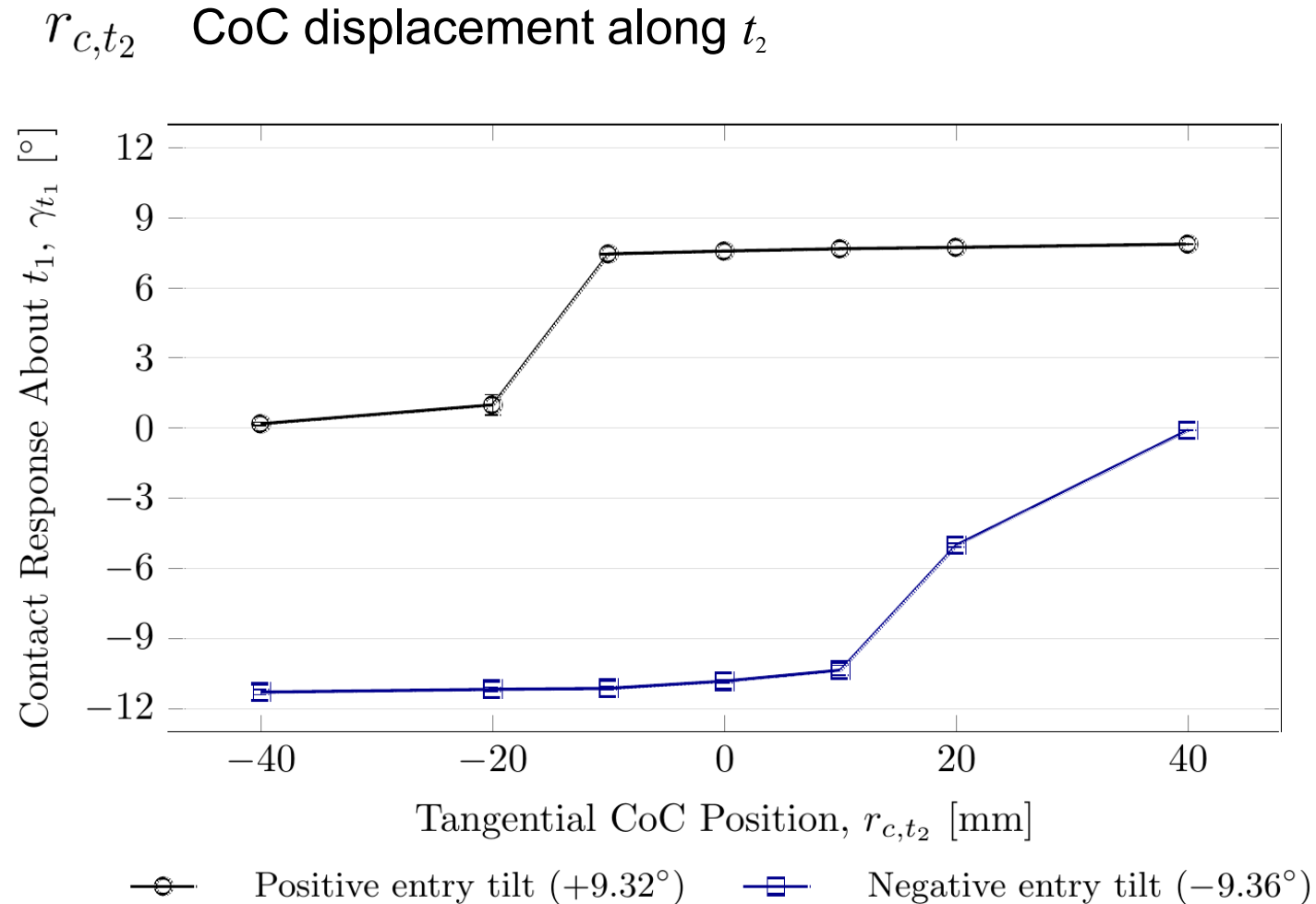


Measured angular offset
 $\theta_{\text{meas},t_1} = (9.24\text{--}9.31)^\circ$

- Rotational stiffness: 63.9% lower response

- Only 0.03° change across the tested range

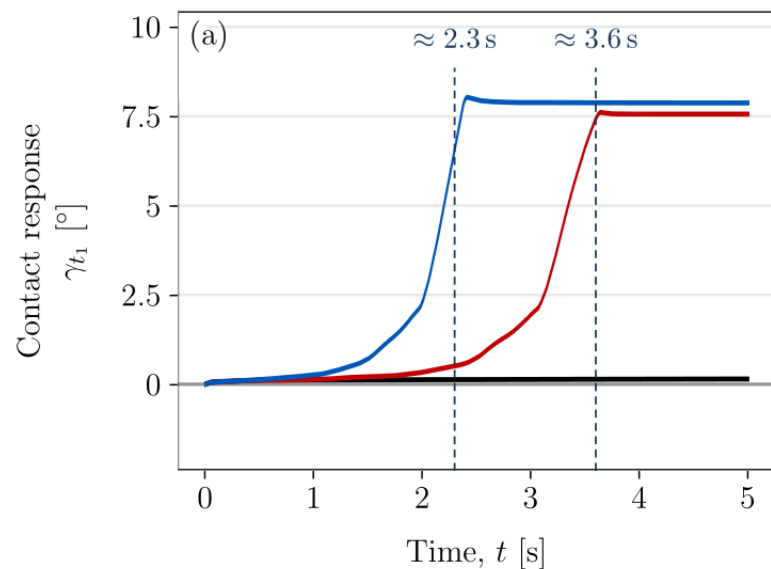
Effect of CoC position



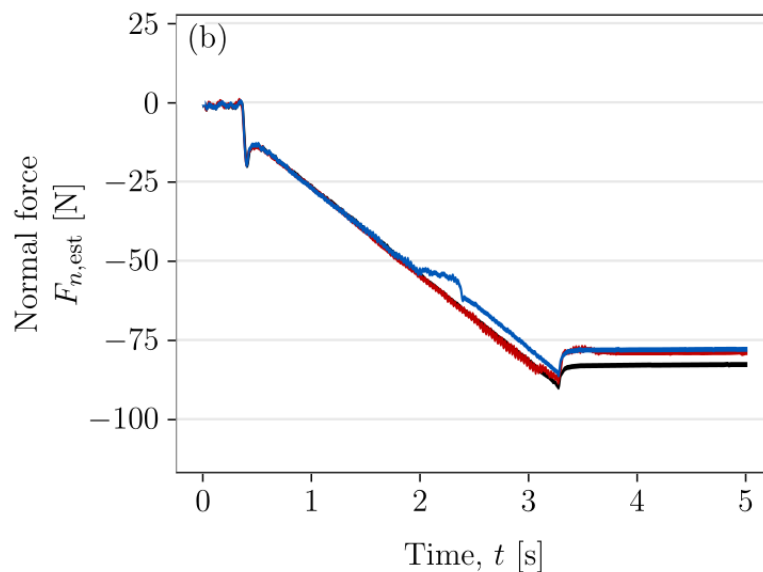
Relative to the TCP-centred baseline

- Supporting displacement: about 4% higher response
- Opposite side: 98–99% lower response
- Required side reverses with the entry tilt

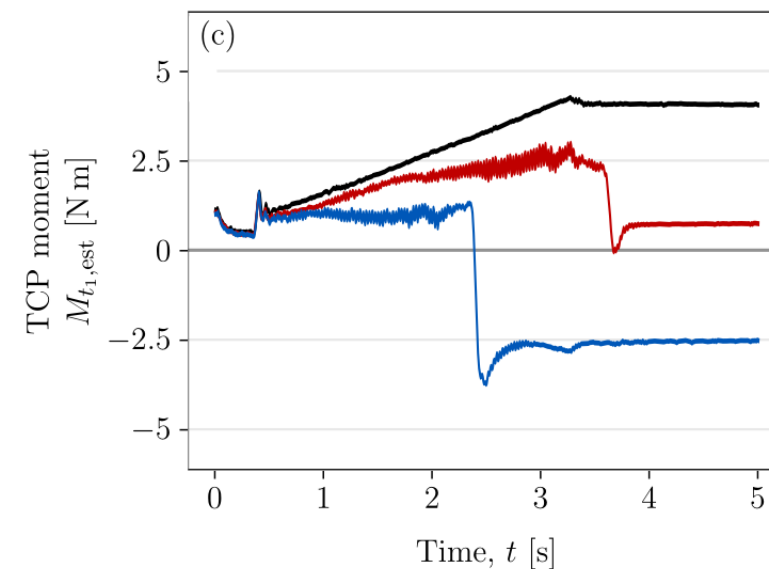
- Contact response



- Normal force

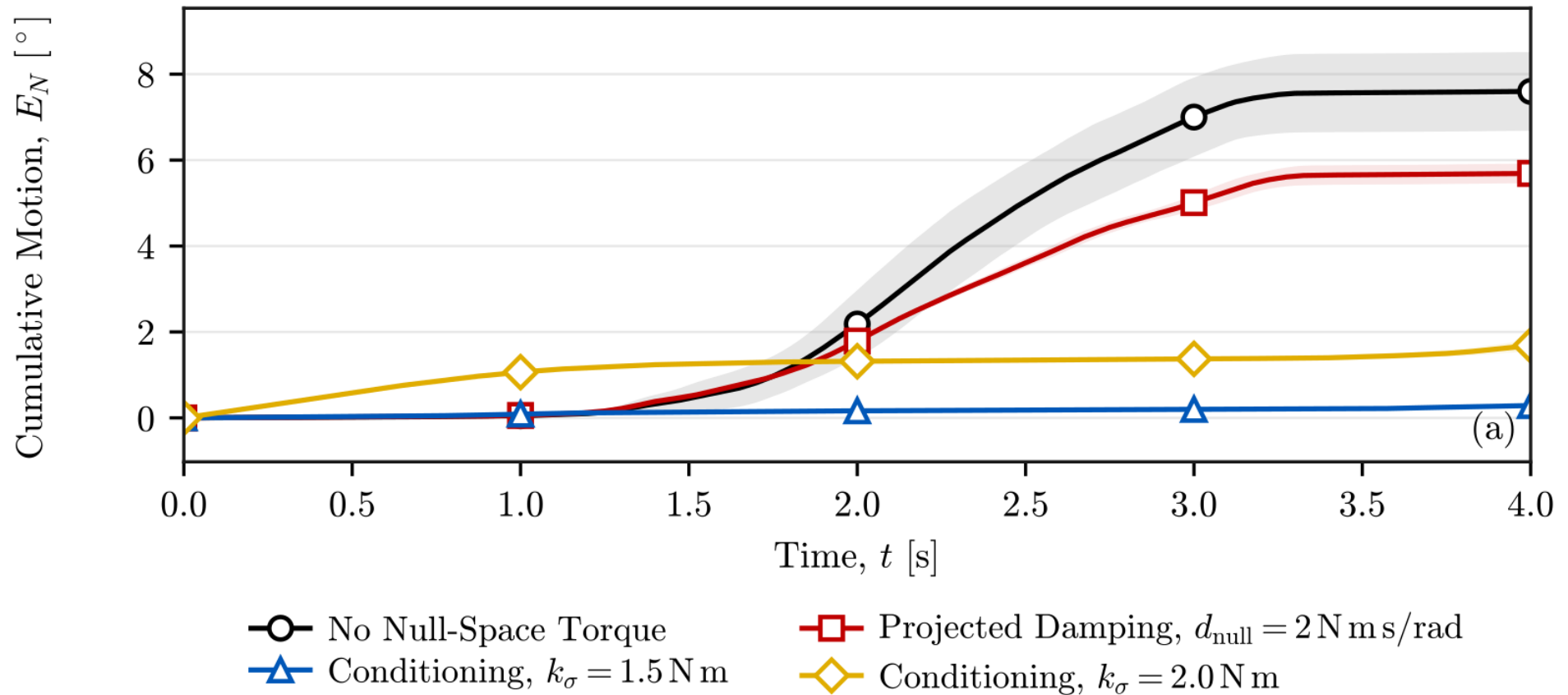


- TCP moment

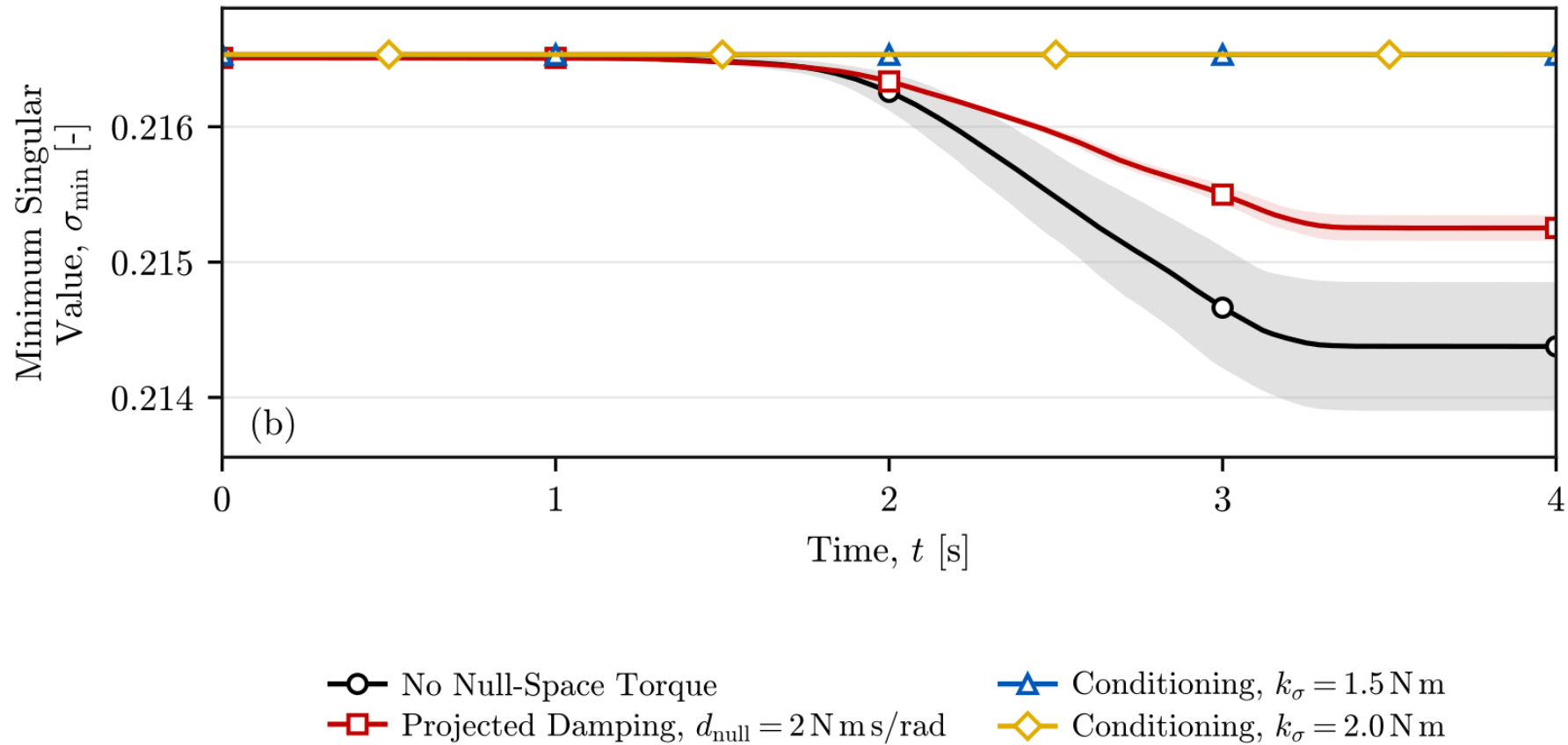


- Reversing CoC displacement reverses the interaction moment
- Supporting displacement: about 1.3 s earlier in these representative traces
- Final normal force (approx.): -83 N (-40 mm), -79 N (TCP), -78 N (+40 mm)

Cumulative null-space motion

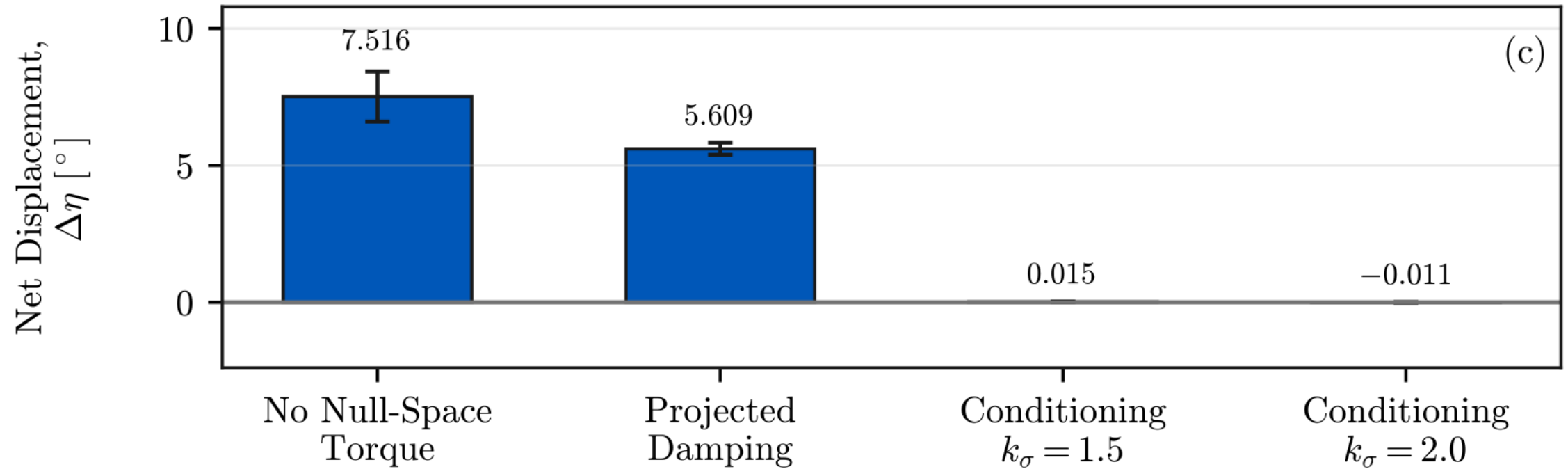


- Damping reduced cumulative motion from 7.598° to 5.689° (25.1%).
- With conditioning, cumulative motion was 0.288° and 1.687° .



- Minimum singular value decreased with no torque and with damping.
- Both conditioning settings kept the indicator nearly constant.

Net null-space displacement



- Conditioning left nearly zero net displacement in this test.
- Small net displacement can coexist with back-and-forth motion.

Conclusion

- Contact-induced rotation towards alignment
- Higher rotational stiffness reduced contact rotation
- Tested tangential stiffness changes had little effect on rotation
- CoC displacement supports or opposes alignment
- Damping reduced motion and conditioning countered the tested disturbance
- Future work
 - More rigid tool mount
 - Adaptive CoC selection and return to the TCP after alignment

- Null-space projector

$$N_\tau = I_7 - J^\top J^{+\top}, \quad \tau_{\text{null}} = \tau_d + \tau_\sigma$$

Projects secondary joint torques into the kinematic null space.

J : joint $\rightarrow$ EE velocities

J^+ : pseudoinverse

I_7 : 7 $\times$ 7 identity matrix

- Damping torque

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}$$

Opposes redundant joint velocity.
Zero at zero joint velocity.

d_{null} : damping coefficient [N m s/rad]

$\dot{q}$: measured joint velocity

- Conditioning torque

$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7)$$

Seeks better local conditioning.
Can act at zero joint velocity.

k_σ : torque magnitude [N m]

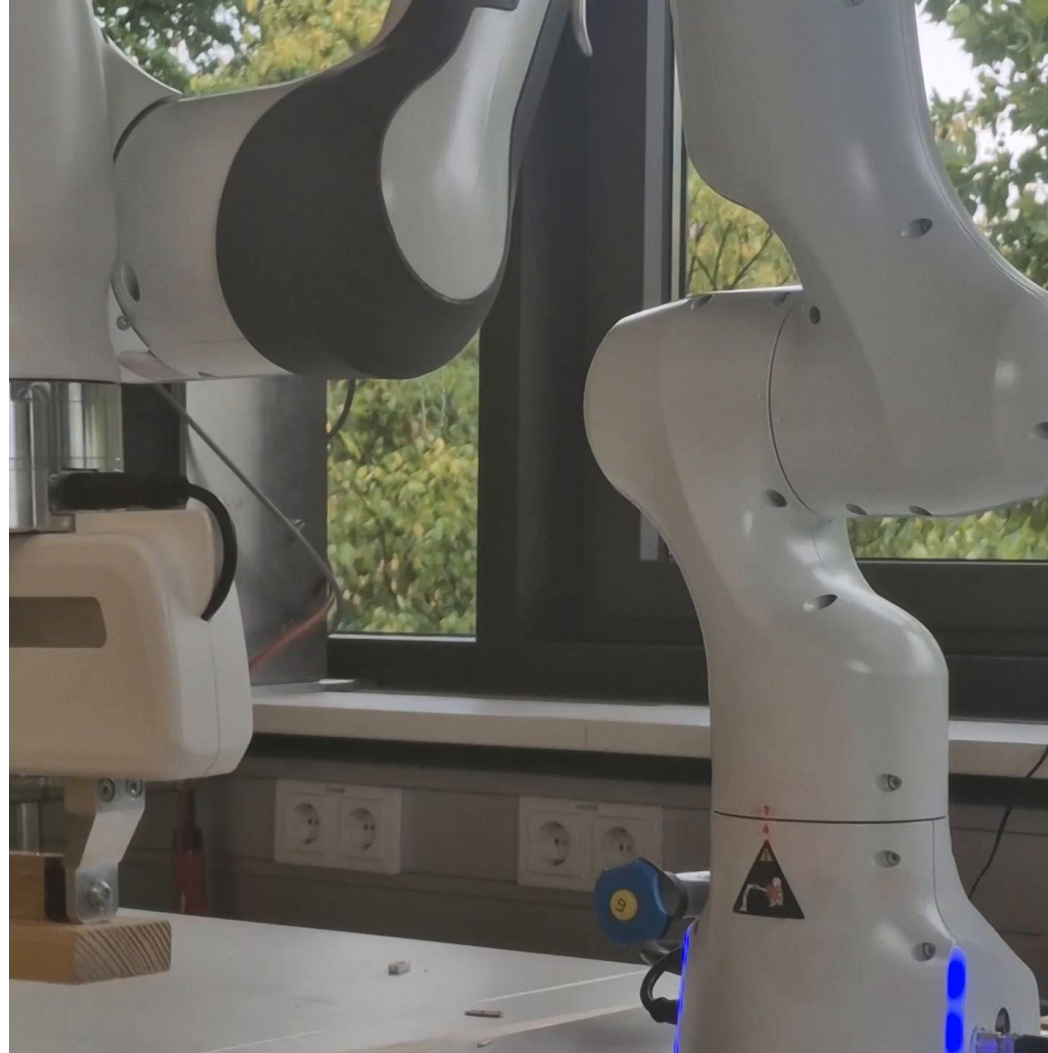
v_7 : unit null-space direction

s_σ : direction choice, zero in deadband



Pre-grinding hold · fixed desired orientation · CoC at TCP

Disturbance demonstration



No null-space torque → Damping → Conditioning